

Scenario

Review the following scenario. Then complete the step-by-step instructions.

You are a level one security operations center (SOC) analyst at a financial services company. You have received an alert about a suspicious file being downloaded on an employee's computer.

You investigate this alert and discover that the employee received an email containing an attachment. The attachment was a password-protected spreadsheet file. The spreadsheet's password was provided in the email. The employee downloaded the file, then entered the password to open the file. When the employee opened the file, a malicious payload was then executed on their computer.

You retrieve the malicious file and create a SHA256 hash of the file. You might recall from a previous course that a **hash function** is an algorithm that produces a code that can't be decrypted. Hashing is a cryptographic method used to uniquely identify malware, acting as the file's unique fingerprint.

Now that you have the file hash, you will use VirusTotal to uncover additional IoCs that are associated with the file.

Has this file been identified as malicious? Explain why or why not.

After accessing the virustotal website , I entered the file's hash into the search bar :



The image shows the VirusTotal search interface. At the top, the VirusTotal logo is displayed in blue. Below the logo, a tagline reads: "Analyse suspicious files, domains, IPs and URLs to detect malware and other breaches, automatically share them with the security community." A navigation bar contains four tabs: "FILE", "URL", "SEARCH", and a settings icon. The "SEARCH" tab is selected and highlighted with a blue underline. Below the navigation bar, there is a magnifying glass icon. Underneath the icon, the text says: "Search for a hash, domain, IP address, URL or gain additional context and threat landscape visibility with OUR THREAT INTELLIGENCE OFFERING." Below this text is a search input field containing the hash "54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b". At the bottom of the form is a "Search" button.

VIRUSTOTAL

Analyse suspicious files, domains, IPs and URLs to detect malware and other breaches, automatically share them with the security community.

FILE URL **SEARCH** ⚙️



Search for a hash, domain, IP address, URL or gain additional context and threat landscape visibility with **OUR THREAT INTELLIGENCE OFFERING.**

54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

Search



Community Score -289

58/72 security vendors flagged this file as malicious

Reanalyze Similar More

54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

bfsvc.exe

Size 430.00 KB

Last Analysis Date 1 day ago



peexe service-scan direct-cpu-clock-access detect-debug-environment spreader runtime-modules long-sleeps checks-user-input

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 30 +

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Contacted URLs (44)

Scanned	Detections	Status	URL
2026-02-02	0 / 94	200	https://fonts.gstatic.com/s/roboto/v18/KFOmCnqEu92Fr1Mu4mxM.woff
2026-02-02	0 / 94	404	https://adservice.google.co.kr/adsid/google/ui?gadsid=AORoGNQnZAiuepi25VY6PFgl8cBBb6AEat1DDBVoE64OR_B59e5p_XMQw
2025-04-27	0 / 97	200	http://o.pki.goog/we2/MFIwUDBOMEwwSjAJBgUrDgMCGGUABBTuM.JxAT2trYla0jia/5EUSmLrk3QQUdb7Ed66J9kQ3fc+xaB8dGuvCNfKCEQDQZgpWpezrXAMFnbj86J49
2026-02-24	12 / 94	-	http://org.misecure.com/index.html
2025-09-03	0 / 98	200	http://ctldl.windowsupdate.com/msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?a98a5653de4a653b
2025-06-24	0 / 97	-	http://www.gstatic.com:443/
2024-10-16	0 / 96	200	https://ssl.gstatic.com/gb/images/i1_1967ca6a.png
2020-10-01	0 / 79	204	https://adservice.google.co.kr/adsid/google/si?gadsid=AORoGNSUAuwYpH0n8JY_v7tQIGWYX2-MeMotUzE8VWqBPXZFsGeh3OoninnP
2022-09-09	0 / 88	200	http://ocsp.pki.goog/gts1o1core/MFIwUDBOMEwwSjAJBgUrDgMCGGUABBRcjDC.Jxnb3nDwij/xz5aZfZjgXvAQUmNH4bhDrz5vsYJ8YkBug630J/SsCEQDOs/DKHslMiAIAAAAAeksR
2025-04-21	0 / 97	200	http://o.pki.goog/we2/MFEwTzBNMEswSTAJBgUrDgMCGGUABBTuM.JxAT2trYla0jia/5EUSmLrk3QQUdb7Ed66J9kQ3fc+xaB8dGuvCNfKCEfbvYhZ XseXDELDPoUP5HuY=

Multiple indicators:

The score of 58/72 indicates that this file is most likely harmful , it also has a community score of -289 and has been categorized as a trojan.

Moreover, it has also tried to contact misecure.com which has been flagged by multiple searchengines.



Community Score -289

58/72 security vendors flagged this file as malicious

Reanalyze Similar More

54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

bfsvc.exe

Size 430.00 KB

Last Analysis Date 1 day ago



peexe service-scan direct-cpu-clock-access detect-debug-environment spreader runtime-modules long-sleeps checks-user-input

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Contacted URLs (44)

Scanned	Detections	Status	URL
2026-02-02	0 / 94	200	https://fonts.gstatic.com/s/roboto/v18/KFOmCnqEu92Fr1Mu4mxM.woff
2026-02-02	0 / 94	404	https://adservice.google.co.kr/adsid/google/ui?gadsid=AORoGNQnZAiuepi25VY6PFgl8cBBb6AEat1DDbVoE64OR_B59e5p_XMQw
2025-04-27	0 / 97	200	http://o.pki.goog/we2/MFlwUDBOMEwwSjAJBgUrDgMCGGUABBTuM.JxAT2trYla0jia/5EUSmLrk3QQUdb7Ed66J9kQ3fc+xaB8dGuvCNfKCEQDQZg pWpezrXAMFnbj86J49
2026-02-24	12 / 94		http://org.misecure.com/index.html
2025-09-03	0 / 98	200	http://ctldl.windowsupdate.com/msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?a98a5653de4a653b
2025-06-24	0 / 97	-	http://www.gstatic.com:443/
2024-10-16	0 / 96	200	https://ssl.gstatic.com/gb/images/i1_1967ca6a.png
2020-10-01	0 / 79	204	https://adservice.google.co.kr/adsid/google/si?gadsid=AORoGNSUAuwYpH0n8JY_v7tQIGWYX2-MeMotUzE8VWqBPXZFgGeh3OoninnP
2022-09-09	0 / 88	200	http://ocsp.pki.goog/gts1o1core/MFlwUDBOMEwwSjAJBgUrDgMCGGUABBCrjDC.Jxnb3nDwj/xz5aZfZjgXvAqUMNH4bhDrz5vsYJ8YkBug630J/SsCEQDOs/DKHsIMiAIAAAAAAeksR
2025-04-21	0 / 97	200	http://o.pki.goog/we2/MFEwTzBNMEswSTAJBgUrDgMCGGUABBTuM.JxAT2trYla0jia/5EUSmLrk3QQUdb7Ed66J9kQ3fc+xaB8dGuvCNfKCEfbvYhZ XseXDELDPoUP5HuY=

Pyramid of Pain values

For the hash we already got it .

We have the org.misecure.com for url and then I used virustotal to check its ip address from his url :



Community
Score

-56

12/93 security vendors flagged this domain as malicious

Reanalyze

More

org.misecure.com

misecure.com

Creation Date

24 years ago

Last Analysis Date

18 hours ago



DETECTION

DETAILS

RELATIONS

COMMUNITY 8

[Join our Community](#) and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Last DNS records

Record type	TTL	Value
A	120	153.234.122.98
AAAA	120	2001:240:2405:da28:4ff0:f8ed:45ed:8be6
CNAME	20	pinhole.dynu.net

Furthermore, by looking at its behaviour , we could see that it made unauthorized changes and file modifications in the OS folder:

Registry actions ⓘ

Registry Keys Opened

- 🔍 HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize
- 🔍 HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize\AppsUseLightTheme
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\AppID\ielowutil.exe
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00020420-0000-0000-C000-000000000046}
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{00020424-0000-0000-C000-000000000046}
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\CLSID\{0002DF01-0000-0000-C000-000000000046}
- 🔍📦 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\Interface\{00000134-0000-0000-C000-000000000046}
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\Interface\{00020400-0000-0000-C000-000000000046}
- 🔍 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\Interface\{3050F1FF-98B5-11CF-BB82-00AA00BDCE0B}
- 🔍📦 HKEY_LOCAL_MACHINE\SOFTWARE\Classes\Interface\{3050F1FF-98B5-11CF-BB82-00AA00BDCE0B}\Forward



TTPs

Phishing,
persistance,evasion

Tools

Mail containing an
attachement and
downlaoded executables

**Network/host
artifacts**

Modification of registry keys
Personalize and AppID.

Domain names

org.misecure.com

IP addresses

153.234.122.98

2001:240:2405:da28:4ff0:f8ed:45ed:8be6

Hash values

54e6ea47eb04634d3e87fd7787e2136ccfbcc80
ade34f246a12cf93bab527f6b